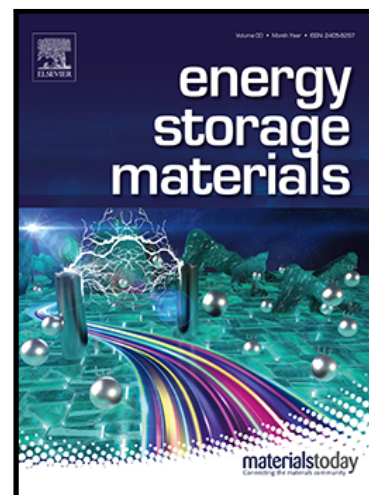


Borophene-like Boron Subunits-inserted Molybdenum Framework of MoB_2 Enables Stable and Quick-acting Li_2S_6 -based Lithium-sulfur Batteries

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Highlights:

- A facile synthesis of MoB₂ materials.
- Borophene-like boron subunits with framework of molybdenum atoms.
- The better performances of lithium-sulfur batteries.

Borophene-like Boron Subunits-inserted Molybdenum Framework of MoB₂ Enables Stable and Quick-acting Li₂S₆-based Lithium-sulfur Batteries

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Rong Wu and Yuwei Zhao co-prepared the material and performed battery measurements under the supervision of Chenyang Zha, Lin Wang and Wei Huang. Chenyang Zha, Lin Wang and Rong Wu co-wrote the manuscript. Huakai Xu carried out theoretical calculations under the supervision of Ouyang Gang. Chengyu Zhang carried out Raman measurements under the supervision of Gang Lu. Rong Wu carried out EIS measurements under the supervision of Tianshi Qin. Chenyang Zha carried out XPS measurements and electrochemical analyses. Jun Deng performed TEM/SEM imaging. Wei Wang, Yao Yin and Chao Zhu assisted in data analysis.

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Abstract

High-performance lithium-sulfur batteries are limited by the severe “shuttle effect” of polysulfide migration. To entrap and immobilize polysulfides, the development of catalytic material is an effective strategy for improving the lithium-sulfur batteries. Herein, we demonstrate that borophene-like boron subunits-inserted molybdenum frameworks of molybdenum diboride (MoB_2) serves as a polysulfide-anchoring center to power redox reaction processing under the high-efficient electron transfer conditions. Specifically, MoB_2 not only offers active sites to anchor polysulfide via covalent B-B and metallic Mo-Mo bonds-based low lithiation structure, but also provides a high conductivity to accelerate polysulfide conversion kinetics. With these advances, the liquid Li_2S_6 -based MoB_2 electrode (area: 2 cm^2) offers a high initial capacity of 1116 mAh/g, and holds 558 mAh/g at 2 C after 500 cycles. Furthermore, the currently proposed MoB_2 catalyst may significantly propel the advancement of electrocatalysis technology from lithium-sulfur batteries to metal-air batteries and carbon dioxide/nitrogen electrochemical reduction.

Keywords

borophene-like boron, molybdenum framework, molybdenum diboride, catalytic ability, electrical conductivity, Li_2S_6 -based batteries

1. Introduction

As the industries of electric vehicles and hand-held electronics develop rapidly, the demand for electrical energy storage with high energy density and long cycle stability grows accordingly.[1-4] Among them, lithium-sulfur (Li-S) batteries are considered as one of the most promising candidates resulting from their high theoretical energy density (2600 Wh/kg) and high theoretical capacity (1675 mAh/g).[1, 2, 5-7] Meanwhile, the sulfur materials also have the advantages of low cost, earth abundance and environmental friendliness. However, there still exist several challenges for practical applications of Li-S batteries which are short cycle life, severe capacity decay and low Coulombic efficiency.[5, 6, 8-10] These poor electrochemical performances come from the well-known “shuttle effect” in which soluble polysulfides generate and diffuse from sulfur cathode to lithium anode, forming a solid passivation layer on lithium anode and ending with the disintegration of the sulfur cathode itself.[1, 3, 5, 6, 8, 9, 11] In addition, the low electronic conductivity of sulfur will cause the sluggish conversion reactions of polysulfides to further accelerate the “shuttle effect”, in turn, the more polysulfides escape into the electrolyte and shuttle between cathode and anode repeatedly. [3, 5, 8, 12, 13]

To address these issues, much research efforts have been devoted to immobilizing the dissolution of polysulfides to promote the polysulfide interconversion.[5, 6] Recently, many functional oxides/sulfides have been explored as the host materials for constraining polysulfides diffusion and catalyzing redox reactions of polysulfides by their interfacial active sites.[5, 6] In particular, TiO_2 and CoS_2 are among the most representative materials owing to their good balance between efficient performance and low cost.[14, 15] For example, X. Lou et al. shows that polar hollow nanospheres of titanium monoxide with highly conductive carbon shells are good polysulfides host. The as-designed composite cathode delivers capacities of 1190 mAh/g at 0.2 C, and maintains stable cell life up to 500 cycles at 0.5 C with a capacity decay rate as low as 0.08 % per cycle.[15] Meanwhile, Q. Zhang et al. demonstrates that sulfiphilic cobalt disulfide not only guarantees effective polarization mitigation and enhances energy efficiency by 10 %, but also promises high cell performance, which shows a high initial capacity of 1368 mAh/g at 0.5 C and stable 2000 cycles with a 0.034 % per cycle of capacity decay rate at 2 C.[14] However, the self-lithiation reaction of host material has always been ignored. To be

specific, Li^+ can insert into or deinserted from the crystal lattice of host materials to cause their structural collapse when polysulfides are anchored in the host materials. In other words, as an ideal catalyst for polysulfides, host materials should have a stable crystal structure with low self-lithiation property, which are beneficial to enhance effective anchoring polysulfides.[5, 6, 16] More importantly, stable crystal structure also strategically optimize for redox electron transfer in the conversion processes of polysulfides to further reduce the “shuttle effect” and enhance redox reaction kinetics. Of note, those synthesis processes of ideal catalyst should be facile and environmentally friendly without hazard gas treating or high-temperature conditions.[7, 17]

Herein, we demonstrate that molybdenum diboride (MoB_2) acts as a novel electrocatalyst in the liquid Li_2S_6 -based high-performance Li-S batteries. The innovation of the MoB_2 materials integrates several unique features: (1) as-designed MoB_2 shows borophene-like boron subunits-inserted molybdenum frameworks of crystal structure, offering abundant active sites to confine polysulfide migration; (2) MoB_2 also reveals a low self-lithiation performance without larger structural collapse when polysulfide is anchored on the MoB_2 surface; (3) as an intrinsic property, MoB_2 shows a high conductivity of $0.15 \times 10^7 \text{ S/m}$ (Pt is $0.9 \times 10^7 \text{ S/m}$) which is significantly beneficial to redox electron transport in the reaction of polysulfide, and further to inhibit their escape from the cathode. As a result, Li-S batteries using MoB_2 -based cathode achieves a high capacity at 2 C with an extremely low fading capacity after 500 cycles.

2. Experimental section

2.1 Material synthesis

All the chemicals were handled in argon filled glovebox. Firstly, MoCl_5 (0.002 mol), boron powder (0.016 mol) and solid Sn powder (0.005 mol) were mixed together. The mixed powder was put in a quartz ampoule, which was sealed at vacuum atmosphere. The ampoule was heated in an tube furnace at 800°C for 4 hours with heating rate of $4^\circ\text{C}/\text{min}$ and cooling naturally. After reaction, the ampoule was opened and the dark product was dispensed into HCl solution (about 10 % mass

concentration). Then as-products were washed several times with deionized water and ethanol. Finally, the MoB_2 was dried in hot air-oven at 120 °C for 12 hours.

2.2 Material characterization

Structures of samples were evaluated employing Powder X-ray diffraction (XRD) analysis (Smartlab Rigaku). Morphologies and microstructures of the samples were examined by field emission scanning electron microscopy (SEM, Gemini instrument operated at 5-15 kV). Transmission electron microscopy (TEM) images were acquired adopting a Tecnai F20 transmission electron microscope (200 kV). In-situ X-ray photoelectron spectroscopy (XPS) measurements were performed on a Thermo Scientific Nexsa XPS system with a monochromatic Al K α radiation ($h\nu = 1486.7$ eV) and a concentric hemispherical analyzer. The Raman measurements were carried out on a confocal Raman spectroscope with an acquisition time of one second. The laser beam was provided by a 532 nm continuous wave laser (LCX-532S, Oxxius), and the power at sample plane was controlled at 10 mW with neutral density filters. The Ultraviolet/Visible adsorption spectra were performed on SHIMADZU UV-1750. Before all in-situ XPS measurements, the batteries were disassembled inside an argon-filled glovebox, and the sample electrodes were carefully rinsed with 3 mL mixed solvents of 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) with a volume ratio of 1:1. The samples were transferred with a specially-built airtight bottle from the glovebox to the spectrometer, to avoid air exposure during the entire sample preparation and transfer processes. All XPS data analysis was carried out with Advantage software.

2.3 Electrochemical characterization

Using a mixture (mass ratio) of 70 % MoB_2 , 20 % conductive carbon and 10 % poly(vinylidene difluoride) in N-methylpyrrolidone as a solvent, as-prepared MoB_2 slurry was casted on the carbon cloth, and vacuum-dried at 120 °C for 24 h. To prepare the 0.5 M Li_2S_6 -containing catholyte, 1.6 g of sulfur powder (S_8), 0.46 g of lithium sulfide (Li_2S) and 0.69 g of lithium nitrate (LiNO_3) were dispersed in 20 mL of dioxolane/dimethoxyethane (DOL/DME, 1:1, v:v) mixed solution with 1 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI). The electrolyte was gently heated at 60 °C for under continuous magnetic stirring to form a homogeneous solution. To prepare

lithium-sulfur batteries in standard CR2032 coin cells, the MoB₂ loaded cathodes were impregnated with a calculated volume of the 0.5 M Li₂S₆-containing catholyte. The average mass of lithium metal is 63 mg (area: 2 cm², thickness: 0.6 mm). For long cycling conditions, 21 μ L Li₂S₆ was added in MoB₂-based electrode (area: 2 cm²) with 18-21 μ L/mg (E/S: electrolyte/sulfur) of electrolyte in which 21 μ L Li₂S₆ for 1 mg/cm² of equivalent sulfur normalized to the cathode area. For the low electrolyte conditions with high sulfur loading, 53 μ L Li₂S₆ was added in MoB₂-based electrode (area: 1 cm²) with 9 μ L/mg of electrolyte in which 53 μ L Li₂S₆ for 5 mg/cm² of equivalent sulfur normalized to the cathode area. It was further paired with a Celgard 2400 polypropylene membrane separator, carbon paper protection of electrode. All the as-assembled cells were activated processes before electrochemical tests. Galvanostatic charge and discharge experiments were controlled by a NEWARE battery tester at various currents in the voltage range of 1.8-2.6 V (versus Li⁺/Li, the same hereafter). Cyclic voltammetry (CV) measurements and chronoamperometric curves were performed on a CHI 660E potentiostat. Electrochemical impedance spectroscopy (EIS) results were collected on a CHI 760E.

2.4 Computational methods and models

The density functional theory (DFT) method was employed to calculate the interactions between different surfaces of molybdenum diboride (MoB₂) and various lithium polysulfide Li₂S_n (n = 8, 6, 4, 2, 1) clusters by using the Vienna Ab initio Simulation Package code (VASP).[18-21] Exchange-correlation potential was described by Perdew-Burk-Ernzerhof (PBE) generalized gradient approximation (GGA) functional.[18-21] The wave function was expressed by plane wave basis with a cutoff energy of 400 eV. The convergence criterion of electron energy and ionic force were adopted with 10⁻⁵ eV and 0.05 eV/Å. Monkhorst-Pack k-point grid with 4 x 3 x 1 and 5 x 3 x 1 were chosen at (100) and (101) surfaces of MoB₂ separately,[21] as for the Li₂S_n clusters, the k-point grid was set to 2 x 2 x 2. The lattice parameters of hexagonal MoB₂ primitive cell with P6/mmm space group were a = b = 0.303 nm and c = 0.333 nm after optimizing, which were consistent with the reported results. In those cases, the 3 x 3 x 3 supercell MoB₂ (100) surface slab and the 4 x 3 x 3 supercell MoB₂ (101) surface slab were chosen as the substrates. There were over 1.5 nm vacuum layers in the z direction of two surface configurations. The outermost

three atom layers had full relaxed in the structural optimization. The binding energy of five possible anode electrode producted including Li_2S_8 , Li_2S_6 , Li_2S_4 , Li_2S_2 and Li_2S on the MoB_2 substrate were estimated. The formula for binding energy was: $E_b = E_{\text{MoB}_2\text{-sur}} + E_{\text{Li}_2\text{S}_n} - E_{\text{MoB}_2\text{-Li}_2\text{S}_n}$ where $E_{\text{MoB}_2\text{-sur}}$, $E_{\text{Li}_2\text{S}_n}$ and $E_{\text{MoB}_2\text{-Li}_2\text{S}_n}$ were the total energies of the MoB_2 surface slab, the Li_2S_n molecular species and the adsorption system, respectively.

3. Results and discussion

More clearly electrochemical redox reactions of Li-ion cells are analyzed by the cyclic voltammetry (CV).[11, 22] In the pure sulfur-based Li-ion cell ($\text{S}_8 // \text{Li}$), sulfur (S_8) is reduced to the long-chain polysulfides around 2.4-2.2 V, and further to short-chain polysulfides, and subsequently to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ around 2.0-1.8 V (red curves in Fig. 1a, c, e). The corresponding charge process shows re-oxidation of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ back to S_8 around 2.1-2.6 V (red curves in Fig. 1a, c, e). In the pure MoB_2 electrode-based Li-ion cell ($\text{MoB}_2 // \text{Li}$), no obvious redox peaks are obtained by CV measurement around 1.5-2.8 V (blue curve in Fig. 1a) which suggests a very low self-lithiation state, and MoB_2 materials can maintain the stable structure without large structural stress (Fig. 1b).[11, 22] That means the MoB_2 can provide an ultra-stable interactive interface for polysulfides in which redox reactions can be sustained processing without material structural change by the Li^+ insertion. As one of the most effective materials, TiO_2 has been widely explored for Li-S batteries, which also has the electrochemical reaction with Li^+ around 1.8-2.6 V (blue curve in Fig. 1c).[12, 15, 23] The Li^+ can insert into layers of anatase TiO_2 (space group $\text{I4}_1/\text{amd}$, No. 141, lattice constants: $a = b = 3.785 \text{ \AA}$, $c = 9.514 \text{ \AA}$) matrix and produce orthorhombic $\text{Li}_{0.5}\text{TiO}_2$ (space group I_{mma} , No. 74, lattice constants: $a = 3.819 \text{ \AA}$, $b = 4.084 \text{ \AA}$, $c = 9.066 \text{ \AA}$), which will cause the distortion of atomic position and change in symmetry of TiO_2 matrix. Specifically, TiO_2 shows $\text{TiO}_2 + 0.5\text{Li}^+ + 0.5\text{e}^- = \text{Li}_{0.5}\text{TiO}_2$ around 1.8-2.2 V where polysulfides also take part in redox reaction around the similar voltage ranges (Fig. 1c).[23, 24] The structural expansion of TiO_2 (or $\text{Li}_{0.5}\text{TiO}_2$) will affect the adsorbed polysulfides on the interfacial active sites, and make the adsorbed polysulfides migration again (Fig. 1d).[5] As another efficient host of polysulfides, CoS_2 exhibits huge structural change to generate Li_2S and Co (blue curve in Fig. 1e).[14] In the discharge process, the crystal structure of CoS_2 is nearly

destroyed by Li^+ intercalation process: $x\text{Li}^+ + xe^- + \text{CoS}_2 \rightarrow \text{Li}_x\text{CoS}_2 + (4 - x) \text{Li}^+ \rightarrow \text{Li}_2\text{S} + \text{Co}$ around 1.8-2.6 V.[14] Compared with CoS_2 and S_8 in the Li ion battery (Fig. 1e), the well-overlapped curves suggest that CoS_2 structure could be broken by Li^+ insertion, where CoS_2 structure has been changed and already difficult to confine polysulfides diffusion (Fig. 1f). Next, the magnified CV curves in the Li ion cells of MoB_2 , TiO_2 and CoS_2 electrodes are shown in Fig. S1a-c (Electronic Supporting Information: ESI). The cycling stability of MoB_2 , TiO_2 and CoS_2 -based Li_2S_6 cells (area: 2 cm^2) with 1 mg/cm^2 sulfur are added in the Fig. S2a. In the 0.5 C current, those MoB_2 , TiO_2 and CoS_2 -based Li_2S_6 cells possess initial discharge capacities of 1288, 930 and 878 mAh/g, and maintain 1194, 786 and 714 mAh/g after 100 cycles, respectively. Moreover, the 50th charge/discharge profiles of the corresponding three electrodes are shown in Fig. S2b. The MoB_2 -based Li_2S_6 cell shows the smallest voltage differences (ΔV), indicating the smallest electrochemical polarization.

In the following, the crystal structure of MoB_2 is characterized by the powder X-ray diffraction (XRD) pattern (Fig. 2a). MoB_2 possesses the hexagonal phase (JCPDS No. 06-0682) and belongs to space group P6/mmm, No. 191 (lattice constants: $a = b = 0.304 \text{ nm}$ and $c = 0.307 \text{ nm}$).[21, 25, 26] Furthermore, the chemical state of the MoB_2 is determined by the X-ray photoelectron spectroscopy (XPS) and these spectra are calibrated by employing the Mo 3d and B 1s peaks. In Fig. 2b, the Mo^{3+} -B and Mo^{6+} -B bindings come from the intrinsic MoB_2 which could imply those multi-active sites could facilitate redox conversion reactions of the different molecule-based polysulfides (S_8 to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$).[25-27] The XPS spectrum of B 1s (Fig. 2c) also exhibits the Mo^{3+} -B and Mo^{6+} -B bonds, and the B-O signal comes from its superficial oxide layer.[25-27] Raman spectra measurements of intrinsic MoB_2 and glass substrate are performed as shown in Fig. 2d. The Raman spectra of MoB_2 at around 226, 272, 324, 365, 649, 810 and 983 cm^{-1} , respectively. As a control sample, the Raman spectra of glass substrate also is given in Fig. 2d. Next, the microstructure of MoB_2 is obtained from scanning electron microscopy (SEM) and high-resolution transmission electron microscopy (HRTEM) images. The unique morphology of MoB_2 nanoparticles with a size of 100-200 nm is presented in Fig. 2e. The HRTEM image (Fig. 2f) reveals a lattice distance of 0.2 nm in reasonably good agreement with the XRD analysis result above of MoB_2 . The selected area electron diffraction (SAED) pattern indicates that the MoB_2 is highly crystallized (inset in Fig. 2f). The highly

crystallized MoB_2 could overcome surface trap/defect to enhance redox reaction kinetics via the effective electron transfer.[5, 7, 17, 28, 29] In the MoB_2 -based hexagonal structure, Mo atoms are metallically bonded into three-dimensional frameworks in which B atoms are covalently bonded into unusual two-dimensional borophene-like B layer that do not exist in isolated forms (Fig. 2g-j).[21, 25-27] To further confirm the relationship between crystalline structure and electrochemical property in MoB_2 materials, the electron location function (ELF) is employed to analyze states of electron cloud distribution in which Mo or B atoms at (001) planes (Fig. 2g-h), and (100)/(101) planes of ELF are given in Fig. 2i-j.[21, 25, 28, 30] It is known that region of ELF close to 1 means the perfect electron localization (lone-pair/covalent electrons), the region of 0.5 represents uniform electron density, and the region near to 0 indicates an electron-deficient state.[21, 25-28, 30] The ELF value of six adjacent B atoms at (001) planes is close to 0.9 (Fig. 2g), which means the existence of a strong B-B covalent chemical bond with sigma type of sp^2 hybridization and similar to borophene. In the Mo-Mo bonding, the correlative ELF values are a range of 0.4 to 0.5, which can distribute all over the (001) planes to represent a uniformly distributed electron cloud-like pure metallic materials (Fig. 2h), and to further show potentially high electron conductance.[21, 25-27] Finally, when electron transfers between Mo/B layers (Fig. 2i-j), most of the area of ELF is around by 0.5 to 0.6, which is beneficial for electrons transfer through such a uniform electron cloud states.[21, 25-27]

Encouraged by these advantages, Li_2S_6 -based MoB_2 cathodes are assessed by battery cycling and rate performances. Those electrodes (area: 2 cm^2 , S loading: 1 mg/cm^2) are measured at a high rate of 0.5 and 1 C in Fig. 3a-b. Those electrodes exhibit a high initial capacity of 1288 and 1233 mAh/g, maintain 710 and 664 mAh/g after 500 cycles with Coulombic efficiency of over 99 %. Another challenge of conventional Li-S cells is a quick-acting cell cycling to meet the practical demands of the energy storage device, which could accelerate the migration of polysulfides and deteriorate the cycle of Li-S cells. In the 2 C current, Li_2S_6 -based MoB_2 electrode (area: 2 cm^2 , S loading: 1 mg/cm^2) reaches 1116 mAh/g, and retains 558 mAh/g after 500 cycles with Coulombic efficiency of approximately 99 % (Fig. 3c). The performances of similar Li-S cells in the literatures are outlined in Table S1. Fig. 3d exhibits the 1st voltage-capacity curves of Li_2S_6 -based MoB_2 electrode at the current

density of 0.5, 1 and 2 C, and the voltage differences (ΔV) are measured at approximately 30 % depth of discharge/charge. Those differences of ΔV are 0.208, 0.241 and 0.327 V under the 0.5, 1 and 2 C, which shows a higher current density does not cause a higher polarization in the Li_2S_6 -based MoB_2 cells (Fig. 3d). Moreover, Li_2S_6 -based MoB_2 cells exhibit discharge capacities of 1288, 1233 and 1116 mAh/g at the 0.5, 1 and 2 C, respectively. The first discharge plateau shows the high-order polysulfide conversion (ΔC_1 : $\text{S}^{2-}_8/\text{S}^{2-}_6$ to S^{2-}_4), and the second discharge plateau shows low-order polysulfide conversion (ΔC_2 : S^{2-}_4 to $\text{S}^{2-}/\text{S}^{2-}_2$). [22, 31] Those Li_2S_6 -based MoB_2 cells show 416, 412 and 383 mAh/g of ΔC_1 , 872, 821 and 733 mAh/g of ΔC_2 at 0.5, 1 and 2 C, respectively (Fig. 3d). In the Fig. 3e, three ratios of $(\Delta C_2)/(\Delta C_1 + \Delta C_2)$ show 67.7 66.6 and 65.7 %, and $\Delta C_2/\Delta C_1$ shows 2.10, 1.99 and 1.91 which both come from the first cycles of Li-S cells. In the ΔC_2 stage, the liquid-solid redox reaction (S^{2-}_4 to $\text{S}^{2-}/\text{S}^{2-}_2$) is relatively sluggish due to phase transition of solid-state diffusion, and the migration of higher-order polysulfides (S^{2-}_4) cannot be avoided. [11, 22, 31, 32] Based on the above analysis, these values of $\Delta C_2/\Delta C_1$ and $\Delta C_2/(\Delta C_1 + \Delta C_2)$ are close to each other by the 0.5 to 2 C, which could prove the stability of MoB_2 electrodes and enhances the electrochemical activity of polysulfide conversion. After 500 cycles (Fig. 3f-g), those ΔV are 0.427, 0.479 and 0.641 V under the 0.5, 1 and 2 C, respectively (Fig. 3f). Electrodes show 225, 215 and 203 mAh/g of ΔC_1 , 485, 449 and 355 mAh/g of ΔC_2 at the 0.5, 1 C and 2 C, respectively, where $\Delta C_2/(\Delta C_1 + \Delta C_2)$ shows 68.3, 67.6 and 63.6 % and $\Delta C_2/\Delta C_1$ shows 2.16, 2.09 and 1.75. Those Li_2S_6 -based MoB_2 electrodes both maintain a continuous contribution to the capacity of ΔC_2 , which shows sufficient redox reaction to restrain a relatively sluggish solid-liquid reaction of polysulfides. [5, 31, 33-35] In the high sulfur mass loading with 9 ul/mg (E/S: electrolyte/sulfur), the Fig. S3 reveals that the MoB_2 -based Li-S cells (5 mg/cm^2) exhibit a high initial capacity of 1064 and 774 mAh/g, maintain 910 and 628 mAh/g after 50 cycles at 0.2 C and 0.5 C, respectively. After the 50 cycles, the SEM images (Fig. S4) of Li metal show smooth surface without obvious cracks and dendrites after the 50 cycles (5 mg/cm^2 under the 0.5 C). What's more, a rate capability is carried out by stepwise ramping current density from 1.0 to 5.0 mA/cm^2 for Li-S batteries using the MoB_2 electrode (Fig. 3h). The cycle capacities are around 1270, 1190, 1060, 945 and 810 mAh/g at 1, 2, 3, 4 and 5 mA/cm^2 , finally the capacity return approximate 1235 mAh/g at 1 mA/cm^2 (Fig. 3h),

which signifies the advantaged performance of MoB₂ with efficient electrochemical kinetics in the Li-S cells. The dependence of the charge/discharge curves are given in Fig. 3i. During the discharge stage, the 1st and 2nd plateaus demonstrate the change from S₈ to polysulfides and to Li₂S₂/Li₂S. During the charge stage, the one plateau reveals the reaction from Li₂S₂/Li₂S to S₈. [11, 22, 36] After rate capability running, those corresponding charge/discharge profiles (Fig. S5) show the high current (5 mA/cm²) charge-discharge running do not reduce the stability of Li-S batteries where the charge/discharge profiles of 1 mA/cm² still overlap well. Fig. S6 shows the optical images of the adsorption test using pure 0.005 M Li₂S₆ and 0.005 M Li₂S₆ with MoB₂. After 52 hours under the 2 °C, turned transparent solution and Ultraviolet/Visible adsorption both show the MoB₂ has a greater capability to adsorb polysulfides due to the strong interactions of active sites.

The electrochemical behavior of the Li₂S₆-based MoB₂ cell is investigated by the cyclic voltammetry (CV) with 1.5-2.8 V at a scan rate of 0.1 mV/s (Fig. 4a) or a modulated scan rate from 0.1 to 1.0 mV/s (Fig. 4c). [7, 8, 11, 22, 24] The CV-based Randles-Sevcik equation could evaluate lithium-ion diffusion coefficient (D_{Li^+}), which is $I_p = 0.4463 n F A C [n F v D / (R T)]^{1/2}$, thereinto, I_p is the current maximum, n is the number of electrons transferred in the redox reaction, F is the Faraday constant, A is the electrode area, C is the concentration, v is the scan rate, D is the diffusion coefficient, R is the gas constant and T is the kelvin temperature. [11, 22] Under 0.1 mV/s, two reduction current peaks are around two reduction voltages (V_{C1} and V_{C2}), which are S₈ to high-order polysulfides and to Li₂S₂/Li₂S. One oxidation peak reveals the one-step reaction around one oxidation voltage (V_A). After 5 cycles of CV running, there are no significant polarization voltages (V_A , V_{C1} and V_{C2}), indicating that the electrochemical reactions of polysulfides are better reversible in the MoB₂ electrode (Fig. 4b). [11, 22] By the rated scanning speed from 0.1 to 1.0 mV/s (Fig. 4c), the increased CV peak currents show a slight shift of peak positions. A linear dependence of the peak current on the square root of the scan rate is determined (Fig. 4d), which could show the stepwise diffusion of Li⁺ for the electrochemical activity of Li₂S₆-based MoB₂ electrode. In addition, those symmetric cells (Fig. 4e) are constructed by MoB₂ with carbon cloth (MoB₂-CC) with loading Li₂S₆ in the electrolyte where a piece of pure carbon cloth (CC) is served as a control in the same conditions. [14, 37, 38] Compared with MoB₂-CC and CC based Li₂S₆ cells in the CV

curves (Fig. 4f), the MoB₂-CC-based cell has higher sensitive current responses, implying the high electrochemical activity of MoB₂ in the redox reaction of polysulfides.[33-35, 39] The chronoamperometric curves (Fig. 4g) clearly disclose the occurrence of diffusion limitation phenomenon in the electrode interface (34 mA/cm² for MoB₂-CC, 11 mA/cm² for CC).[40] As shown by the straight line passing through the origin in the Cottrell plot (Fig. 4g), the MoB₂-CC-based cell has a higher current response than the CC-based cell, which suggests a better interaction between the Li₂S₆ and MoB₂ to power redox electrochemical reaction processing.[40] The electrochemical impedance spectroscopy (EIS) indicates that the reaction semicircle of the MoB₂-CC-based cell has a similar diameter with the CC-based cell, suggesting the minor difference of the charge transfer resistance for the two cells (Fig. 4h).[11] Those above-mentioned results are summarized by the relationship between the polysulfide affinity and conductivity improvement of MoB₂ in the polysulfide redox reactions without the influence of Li metal.

To further assess favorable factors for the electrochemical performance of Li₂S₆-based MoB₂ electrodes, in-situ XPS technology is used to examine the interaction between polysulfides and cathode interface.[7, 8, 12, 41] The in-situ XPS fine spectra (Fig. 5a-d) present the interfacial chemical environments and bonding configurations in electrodes for two different stages: (1) the cell ran 10 times (10th) after aged time, and (2) the cell ran 500 times (500th) after aged time. Fig. 5a presents eight characteristic peaks of Mo 3d and one characteristic peak of S 2s around 240-222 eV. Results of the Mo⁶⁺-S and Mo³⁺-S binding are found during 500 cycles, which could be an available channel for electron migration and enhance redox reactions of polysulfides on MoB₂ surfaces to inhibit the “shuttle effect”. [7, 8, 12, 41] The peaks of Mo⁶⁺-B and Mo³⁺-B peaks reveal Mo in the intrinsic MoB₂. In the spectra of XPS B 1 s (Fig. 5b), the O-B-Li, Mo⁶⁺-B-Li and Mo³⁺-B-Li peaks could be attributed to the Li⁺ effect. The S 2p has six characteristic peaks from 172 to 160 eV (Fig. 5c), which corresponds to LiTFSI salt (bis-trifluoromethane sulfonamide lithium salt), Mo-S and polysulfide intermediates (Li-S_x) bonding on MoB₂ surfaces. Peaks of Mo-S bonding are consistent with the XPS results of Mo 3d. Peaks of the LiTFSI salt and Li-S_x could be ascribed to sulfur-containing bonds (S-O, S-N...) and/or residual sulfur-containing materials.[7, 8, 41, 42] In the spectra of XPS C 1 s (Fig. 5d), those C-O/C-C/C-S peaks could be attributed to the carbon-oxygen group of 1,3-dioxolane

(DOL), 1,2-dimethoxyethane (DME), bis(trifluoromethane) sulfonamide lithium salt (LiTFSI). The position of LiTFSI peak in the C 1s spectrum consisted of mono-fluorinated carbon (C-F or -CHF), which could be from the product fluoroethylene carbonate, and such these uniformly distributed F-containing species are across the electrode interface.[41, 42]

To gain further insight into the role of MoB₂ materials along with these mechanisms in the Li-S cell, the first principle-based theoretical calculation is investigated to an interaction between experimental and theoretical results.[18-21, 43] Fig. 6a shows side and top views of MoB₂ (101) facets slabs, which is more commonly obtained. For comparison, the MoB₂ (100) substrate is also studied (Fig. S7). Fig. 6b-f and Fig. S7b-f show the binding energies between various polysulfides and MoB₂ (101) and (100) facets. The adsorption energies of Li₂S₈ (-7.7 eV), Li₂S₆ (-7.2 eV), Li₂S₄ (-11.1 eV), Li₂S₂ (-8.4 eV) and Li₂S (-7.7 eV) on MoB₂ (101) were stronger than the corresponding ones on MoB₂ (100) facets (-5.6 eV for Li₂S₈, -5.7 eV for Li₂S₆, -6.7 eV for Li₂S₄, -6.4 eV for Li₂S₂ and -5.9 eV for Li₂S). The surface energy for the MoB₂ (100) and (101) surfaces are 0.43 and 0.48 eV/Å², resulting in the higher polysulfide binding energy for (101) surface ranging from 7.2 to 11.1 eV. Also, two surface configurations broadly increase firstly and then decrease when the molecular length of cluster decreases from Li₂S₈ to Li₂S which could indicate a stable discharge platform with relatively low 2.1 eV and will be observed during cycling. Therefore, MoB₂ suppresses the “shuttle effect” and greatly improves the utilization of Li⁺. Furthermore, the Li atom loses the electrons and the S atom gains the electrons of Li₂S on the MoB₂ surfaces. However, S atoms are only a small proportion of clusters, accounting for electrons transfer from the cluster to the surface. Instead, the charge tends to shift from substrate to molecule for the Li₂S₄ or larger cluster, showing the polarity reversal occurs in the polarized adsorption system. It is noteworthy that some S-S bonds in the long-chain cluster break, but new S-B or S-Li bonds do not form, leading to the binding energy reduces in the adsorption process. Furthermore, Bader charge analysis is used for quantitatively describing the charge transfer between clusters and surfaces. It should be noted that if the value is positive (negative), the clusters obtain (loss) electron. Bader charges of polysulfide are 0.47e for Li₂S₈, 0.73e for Li₂S₆, 0.74e for Li₂S₄, -0.48e for Li₂S₂ and -0.97e for Li₂S on the MoB₂ (101) facets. Also, Bader charges of polysulfide on the MoB₂ (100) facets are

1.61e for Li_2S_8 , 0.73e for Li_2S_6 , 1.66e for Li_2S_4 , 0.72e for Li_2S_2 and -0.54e for Li_2S . Compared to the (100) surface, the clusters on (101) surface are easier to obtain electrons, which are beneficial for the lithiation process of sulfur. A conclusion thus can be drawn that MoB_2 shows strong adsorption ability to polysulfide, and can be a promising material for Li-S battery. Partial density of states (PDOS) in Fig. 6g-i, taking Li_2S_8 , Li_2S_4 and Li_2S for examples, illustrate that all configurations remain metal behavior ensuring high conductivity of anode. In the long chain polysulfide absorption system (Li_2S_8 and Li_2S_4), lots of overlap peaks between S-p and B-p states are found in the energy region about -5 eV, which is mainly contributed by strong hybridization between S and B atoms. In contrast, the S-Mo hybridization is weaker due to more B atoms on the facets. Comparing to the case of (100) surface (Fig. S7g-i), the PDOS peak of S atom in Li_2S_n on (101) surface is closer to the Fermi level, meaning that the sulfur atom has higher chemical activity.

4. Conclusion

In summary, we reported the preparation of highly crystallized MoB_2 nanoparticles, and investigated their application as the cathode electrocatalyst for Li-S batteries. Benefit from the borophene-like boron layer and metallic Mo-based three dimensional frameworks of structure, MoB_2 not only powers the polysulfides conversion in the low self-lithiation state, but also improves reaction kinetics of polysulfides via a better electronic conductivity. The Li_2S_6 -based MoB_2 cathode exhibits large specific capacity, impressive rate capability and excellent cycling stability even at high sulfur loading. More importantly, our results present a novel impetus to the tailored design of borophene-based metal materials with high catalytic performances for the future electrocatalysis field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at:

Figure

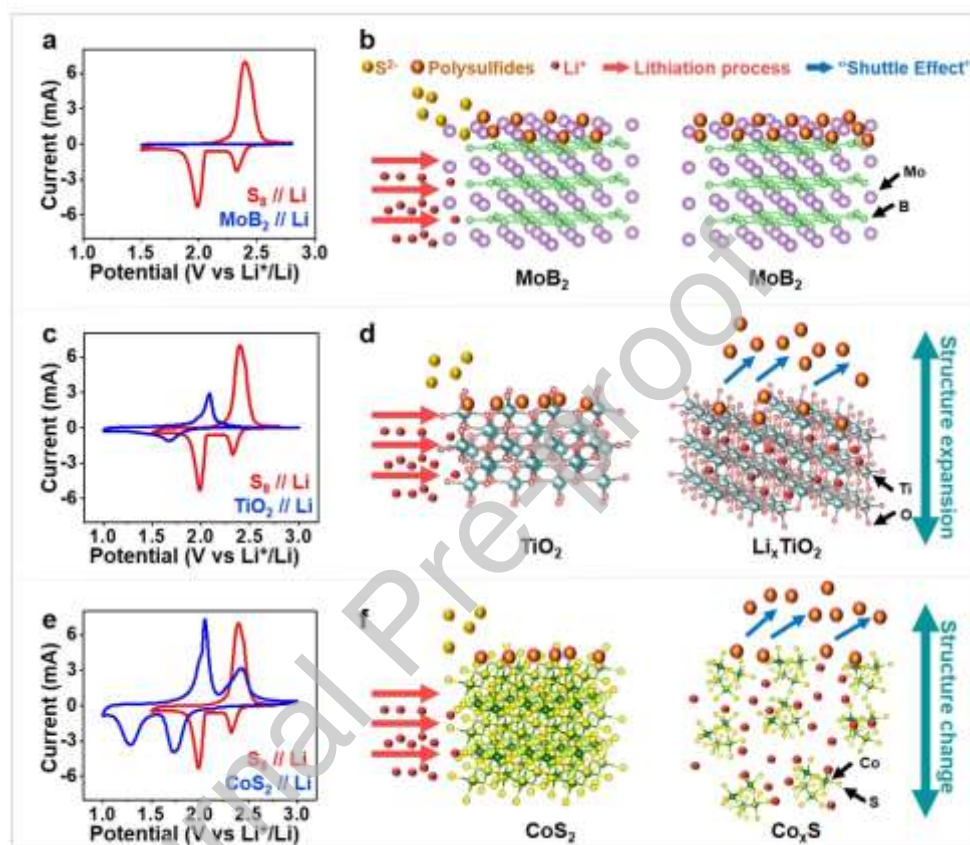


Fig. 1. CV curves in the Li ion cells and schematic illustration showing the self-lithiation process in polysulfide-containing electrolyte of MoB₂ (a-b), TiO₂ (c-d) and CoS₂ (e-f), respectively. CV curves (red line) of pure S₈ cathode as the controlled electrode in the Li ion cells.

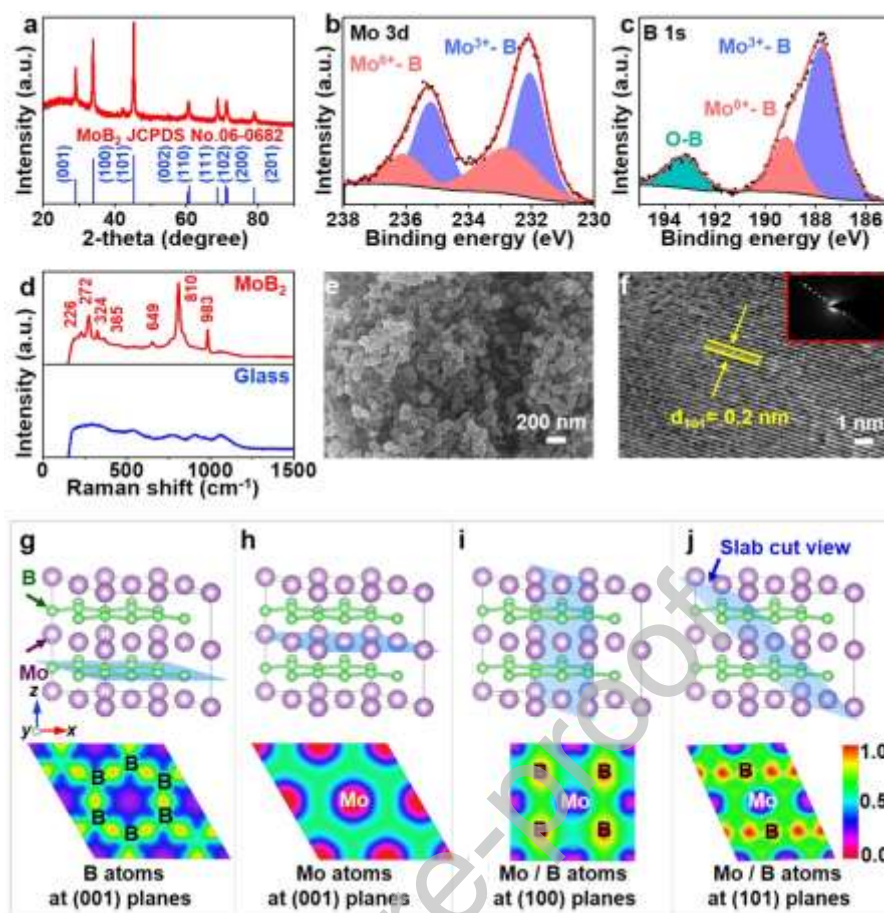


Fig. 2. Structural characterizations of MoB₂. XRD pattern (a) of MoB₂. XPS spectra of Mo 3d (b) and B 1s (c) in the MoB₂. Raman spectra (d) of MoB₂ and glass substrate. SEM image (e) and TEM image (f) with an inset of SAED pattern of MoB₂. Crystal structure (top) and electron location function (bottom) for MoB₂ with B atoms at (001) planes (g), Mo atoms at (001) planes (h), Mo and B atoms at (100) planes (g), Mo and B atoms at (101) planes (j), respectively.

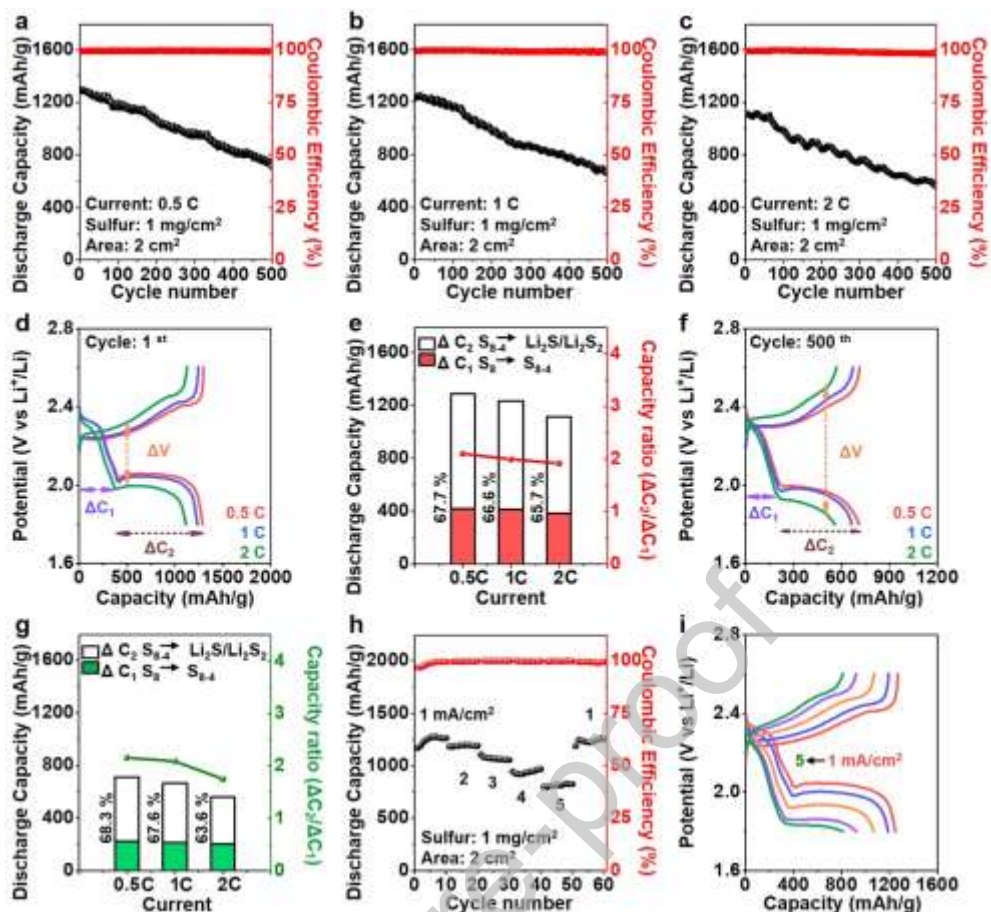


Fig. 3. Electrochemical performance of Li_2S_6 -based MoB_2 cells. The cycling stability and Coulombic efficiency of Li-S cells under different current conditions from 0.5 to 2 C (a-c). The charge and discharge profiles (d) from 0.5 to 2 C at 1st cycle. Ratios image (e) of $\Delta C_2/\Delta C_1$ for MoB_2 cathodes. The charge and discharge profiles (f) from 0.5 to 2 C at 500th cycle. Ratios image (g) of $\Delta C_2/\Delta C_1$ for MoB_2 cathodes. Rate performances and Coulombic efficiency of cell from 1 to 5 mA/cm² (h) with corresponding charge and discharge profiles (i).

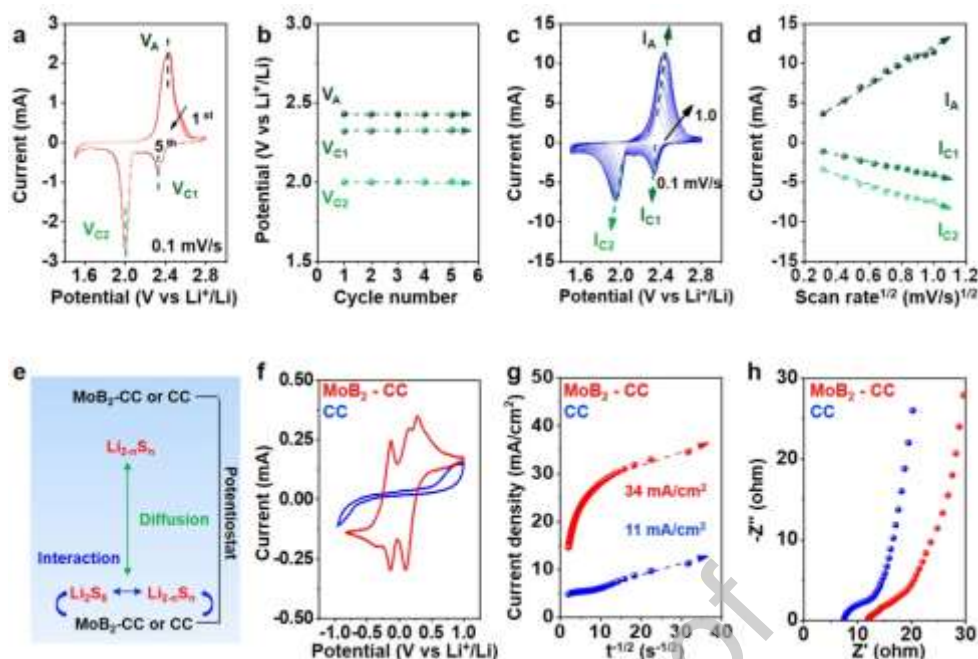


Fig. 4. Stabilized CV curves (a) for a Li_2S_6 -based MoB_2 cell scanned from the 1st to the 5th times at 0.1 mV/s with the corresponding cathodic and anodic peak potentials from the cycle times showing a linear correlation (b). Rated CV curves from 0.1 to 1.0 mV/s (c) with the corresponding cathodic and anodic peak currents from the square root of the scan rate showing a linear correlation (d). Schematic illustration of a Li_2S_6 -based symmetric cell (e). CV curves (f), Cottrell plots of the chronoamperometric curves (g), and Nyquist plots (h) of the MoB_2 -CC and CC-based electrodes.

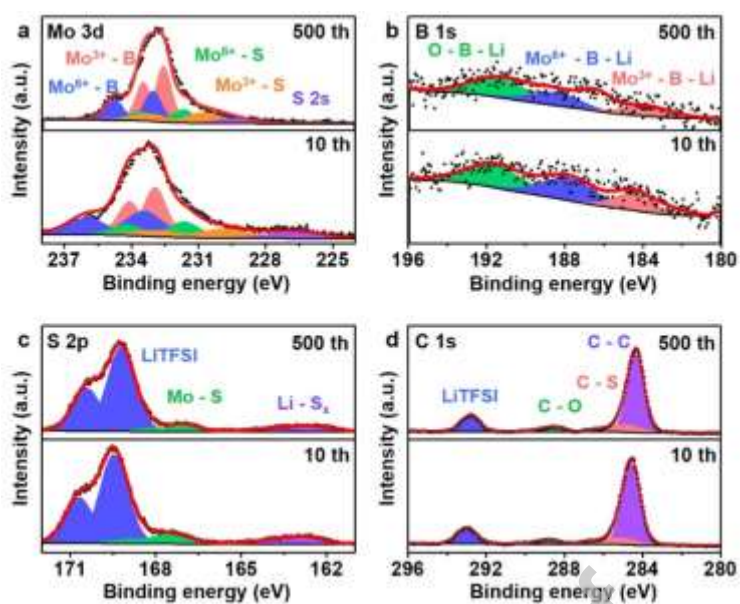


Fig. 5. The Mo 3d (a), B 1s (b), S 2p (c) and C 1s spectra (d) of in-situ XPS from the Li_2S_6 -based MoB_2 cells after the 10th and 500th cycles. The measured curves are in black, and the fitted ones are in red.

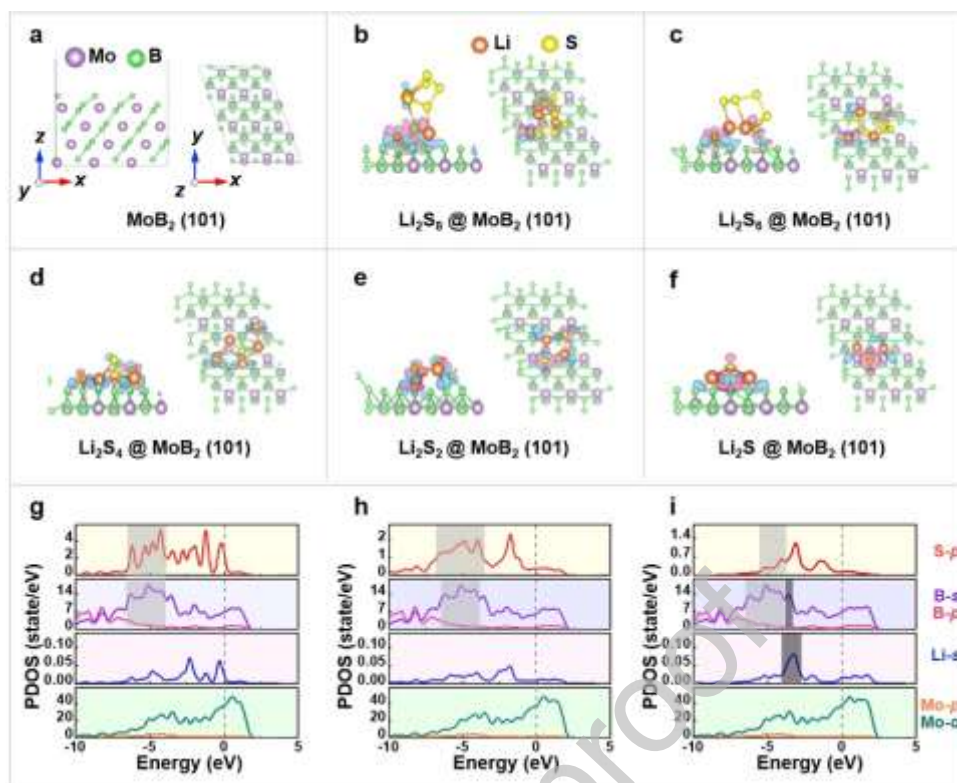


Fig. 6. Side and top views of MoB_2 (101) facets slabs (a). Side and top views of charge density difference for the most stable adsorption configurations of Li_2S_8 , Li_2S_6 , Li_2S_4 , Li_2S_2 and Li_2S molecules on MoB_2 (101) facets. The blue and pink areas are the regions of trapped and lost charge, respectively (b-f). PDOSs of Li_2S_8 , Li_2S_4 and Li_2S adsorb on the MoB_2 (101) facets (g-i), and the S-B and Li-B hybridizations are marked with gray area.

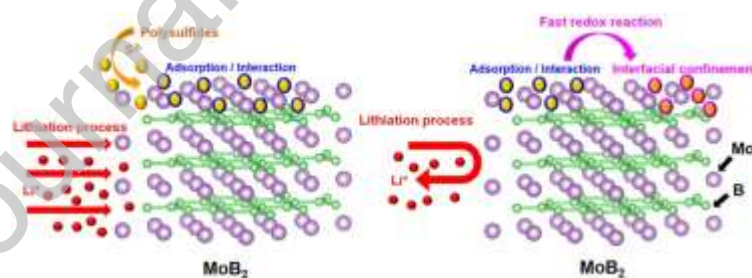
References

- [1] M. Wild, L. O. Neill, T. Zhang, R. Purkayastha, G. Minton, M. Marinescu, G.J. Offer, *Energy Environ. Sci.*, 8 (2015) 3477-3494.
- [2] J.B. Goodenough, *Energy Storage Mater.*, 1 (2015) 158-161.
- [3] P.G. Bruce, S.A. Freunberger, L.J. Hardwick, J. Tarascon, *Nat. Mater.*, 11 (2012) 19-29.
- [4] Q. Yun, L. Li, Z. Hu, Q. Lu, B. Chen, H. Zhang, *Adv. Mater.*, 32 (2020) 1903826.
- [5] M. Zhang, W. Chen, L. Xue, Y. Jiao, T. Lei, J. Chu, J. Huang, C. Gong, C. Yan, Y. Yan, Y. Hu, X. Wang, J. Xiong, *Adv. Energy Mater.*, 10 (2020) 1903008.
- [6] D. Liu, C. Zhang, G. Zhou, W. Lv, G. Ling, L. Zhi, Q.H. Yang, *Adv. Sci.*, 5 (2018) 1700270.
- [7] C. Zha, X. Zhu, J. Deng, Y. Zhou, Y. Li, J. Chen, P. Ding, Y. Hu, Y. Li, H. Chen, *Energy Storage Mater.*, 26 (2020) 40-45.
- [8] C. Zha, D. Wu, T. Zhang, J. Wu, H. Chen, *Energy Storage Mater.*, 17 (2019) 118-125.
- [9] Y.V. Mikhaylik, J.R. Akridge, *J. Electrochem. Soc.*, 151 (2004) A1969-A1976.
- [10] C. Zha, D. Wu, Y. Zhao, J. Deng, J. Wu, R. Wu, M. Yang, L. Wang, H. Chen, J. *Energy Chem.*, 52 (2021) 163-169.
- [11] X. Yang, A.L. Rogach, *Adv. Energy Mater.*, 9 (2019) 1900747.
- [12] C. Zha, X. Gu, D. Wu, H. Chen, *J. Mater. Chem. A*, 7 (2019) 6431-6438.
- [13] C. Zha, F. Yang, J. Zhang, T. Zhang, S. Dong, H. Chen, *J. Mater. Chem. A*, (2018) 16574-16582.
- [14] Z. Yuan, H. Peng, T. Hou, J. Huang, C. Chen, D. Wang, X. Cheng, F. Wei, Q. Zhang, *Nano Lett.*, 16 (2016) 519-527.
- [15] Z. Li, J. Zhang, B. Guan, D. Wang, L. Liu, X.W. Lou, *Nat. Commun.*, 7 (2016) 13065.
- [16] C. Wu, H. Huang, W. Lu, Z. Wei, X. Ni, F. Sun, P. Qing, Z. Liu, J. Ma, W. Wei, *Adv. Sci.*, (2020) 1902643.
- [17] Z. Li, Q. He, X. Xu, Y. Zhao, X. Liu, C. Zhou, D. Ai, L. Xia, L. Mai, *Adv. Mater.*, 30 (2018) 1804089.
- [18] P.E. Blöchl, *Phys. Rev. B*, 50 (1994) 17953.
- [19] G. Kresse, J. Furthmüller, *Phys. Rev. B*, 54 (1996) 11169.

- [20] J.P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.*, 77 (1996) 3865.
- [21] L. Ding, P. Shao, F. Zhang, C. Lu, L. Ding, S.Y. Ning, X.F. Huang, *Inorg. Chem.*, 55 (2016) 7033-7040.
- [22] H. Ye, J. Sun, S. Zhang, H. Lin, T. Zhang, Q. Yao, J.Y. Lee, *ACS Nano*, 13 (2019) 14208-14216.
- [23] C. Zha, D. He, J. Zou, L. Shen, X. Zhang, Y. Wang, H. Kung, N. Bao, *J. Mater. Chem. A*, 2 (2014) 16931-16938.
- [24] T. Lei, Y. Xie, X. Wang, S. Miao, J. Xiong, C. Yan, *Small*, 13 (2017) 1701013.
- [25] Y. Chen, G. Yu, W. Chen, Y. Liu, G.-D. Li, P. Zhu, Q. Tao, Q. Li, J. Liu, X. Shen, *J. Am. Chem. Soc.*, 139 (2017) 12370-12373.
- [26] H. Park, Y. Zhang, J.P. Scheifers, P.R. Jothi, A. Encinas, B.P. Fokwa, *J. Am. Chem. Soc.*, 139 (2017) 12915-12918.
- [27] H. Park, A. Encinas, J.P. Scheifers, Y. Zhang, B.P. Fokwa, *Angew. Chem., Int. Ed.*, 56 (2017) 5575-5578.
- [28] C. Zha, J. Zhang, L. Shen, S. Dong, A. Gupta, N. Bao, *Chemistryselect*, 2 (2017) 10971-10978.
- [29] C. Zha, C. Pang, Y. Xu, C. Ji, X. Zhang, L. Shen, Y. Wang, A. Gupta, N. Bao, *RSC Adv.*, 6 (2016) 62226-62235.
- [30] C. Zha, C. Ji, J. Zhang, L. Shen, X. Zhang, S. Dong, N. Bao, *RSC Adv.*, 6 (2016) 107151-107157.
- [31] D. Gueon, J.T. Hwang, S.B. Yang, E. Cho, K. Sohn, D.-K. Yang, J.H. Moon, *ACS Nano*, 12 (2018) 226-233.
- [32] H. Yuan, H. Peng, B. Li, J. Xie, L. Kong, M. Zhao, X. Chen, J. Huang, Q. Zhang, *Adv. Energy Mater.*, 9 (2019) 1802768.
- [33] Y. Li, C. Wang, W. Wang, A.Y.S. Eng, M. Wan, L. Fu, E. Mao, G. Li, J. Tang, Z.W. Seh, *ACS Nano*, 14 (2019) 1148-1157.
- [34] J. Ma, N. Li, Q. Zhang, X. Zhang, J. Wang, K. Li, X. Hao, J. Yan, *Energy Environ. Sci.*, 11 (2018) 2833-2838.
- [35] X. Zhang, W. Wang, A. Wang, Y. Huang, K. Yuan, Z. Yu, J. Qiu, Y. Yang, *J. Mater. Chem. A*, 2 (2014) 11660-11665.
- [36] H. Lin, L. Yang, X. Jiang, G. Li, T. Zhang, Q. Yao, G.W. Zheng, J.Y. Lee, *Energy Environ. Sci.*, 10 (2017) 1476-1486.

- [37] J. Pu, Z. Shen, J. Zheng, W. Wu, C. Zhu, Q. Zhou, H. Zhang, F. Pan, *Nano Energy*, 37 (2017) 7-14.
- [38] J. Xu, W. Zhang, H. Fan, F. Cheng, D. Su, G. Wang, *Nano Energy*, 51 (2018) 73-82.
- [39] P.R. Jothi, Y. Zhang, J.P. Scheifers, H. Park, B.P. Fokwa, *Sustainable Energy Fuels*, 1 (2017) 1928-1934.
- [40] Y. Kato, S. Hori, T. Saito, K. Suzuki, M. Hirayama, A. Mitsui, M. Yonemura, H. Iba, R. Kanno, *Nat. Energy*, 1 (2016) 16030.
- [41] F. Lindgren, C. Xu, L. Niedzicki, M. Marcinek, T. Gustafsson, F. Bjorefors, K. Edstrom, R. Younesi, *ACS Appl. Mater. Interfaces*, 8 (2016) 15758-15766.
- [42] F. Lindgren, C. Xu, J. Maibach, A.M. Andersson, M. Marcinek, L. Niedzicki, T. Gustafsson, F. Bjorefors, K. Edstrom, *J. Power Sources*, 301 (2016) 105-112.
- [43] L. Zhang, P. Liang, H. Shu, X. Man, F. Li, J. Huang, Q. Dong, D. Chao, *J. Phys. Chem. C*, 121 (2017) 15549-15555.

Graphical abstract:



Borophene-like boron subunits-inserted molybdenum frameworks of MoB₂ serves as a polysulfide-anchoring center to power redox reaction processing.